

Topology of Deep Neural Networks

Overview

- ReLU fulfills the necessity for a non-injective, topologically invariant activation function
- Topological manifolds have uncountably many points in their sets, but in reality, we use a finite subset of the manifold and use persistent homology as a tool to give meaning to the finite subset and estimate the Betti numbers of the underlying manifold
- Footnote 2: “codes available for the reader’s further experimentation”
- Homeomorphic mappings preserve topology; nonhomeomorphic mappings, such as ReLU, can change topology
- In practice, homeomorphic mappings achieve topology changes through round-off errors
- We use $x \mapsto \max(Ax + b, 0)$, where $Ax + b$ moves any “topologically complicated” part of M outside the nonnegative orthant and ReLU collapses these into a single point

Quantifying Topology

- Betti numbers, $\beta = (\beta_0, \beta_1, \beta_2, \dots)$: count the number of k -dimensional holes of a manifold; loosely:
 - 0-dimensional holes: the number of connected components of a manifold
 - 1-dimensional holes: the number of independent loops that cannot be continuously shrunk to a point
 - 2-dimensional holes: the number of independent closed surfaces that enclose a 3-dimensional void
- Topological complexity: $\omega(M) = \beta_0(M) + \beta_1(M) + \beta_2(M) + \dots$
- Our goal is to track the Betti numbers

$$\beta(M) \rightarrow \beta(\nu_1(M)) \rightarrow \beta(\nu_2(M)) \rightarrow \dots \rightarrow \beta(\nu_{l-1}(M)) \rightarrow \beta(\nu_l(M)).$$

To do this in reality, we will have to estimate $\beta(M)$ from a point cloud, $X \subset M$, sampled from M , possibly with noise

Algebraic Topology and Persistent Homology Background

Simplicial Complexes

- Simplex, σ : (k -dimensional) the convex hull of $k + 1$ affinely independent points $v_0, \dots, v_k \in \mathbb{R}^d$
 - 0-dimensional: a point
 - 1-dimensional: a line segment
 - 2-dimensional: a triangle
 - 3-dimensional: a tetrahedron
- Simplicial Complex, K : (m -dimensional) a set of simplices of at most dimension m , satisfying the following:
 - (i) $\sigma_0 \subset \sigma \in K \implies \sigma_0 \in K$
 - (ii) $\sigma_1, \sigma_2 \in K, \sigma := \sigma_1 \cap \sigma_2 \neq \emptyset \implies \sigma \subset \sigma_1$ and $\sigma \subset \sigma_2$

Homology and Betti Numbers

$$0 \xrightarrow{\partial_{d+1}} C_d \xrightarrow{\partial_d} C_{d-1} \xrightarrow{\partial_{d-1}} \cdots \xrightarrow{\partial_{k+1}} C_k \xrightarrow{\partial_k} C_{k-1} \xrightarrow{\partial_{k+1}} \cdots \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0 \xrightarrow{\partial_0} 0$$

Let $M \subset \mathbb{R}^d$ be a manifold and $k \in \{0, 1, \dots, d\}$:

- The k -th homology group of M is given by $H_k = \ker(\partial_k)/\text{im}(\partial_{k+1})$
- Homology classes are elements of H_k which form cosets/equivalence classes
- Then, $\dim(H_k) = \beta_k$, the k -th Betti number of M

Simplicial Homology

- Given an abstract simplicial complex, we define an $\mathbb{F}_2$ -vector space $C_k(K)$ as follows: Let $K^{(k)} = \{\sigma_1, \sigma_2, \dots, \sigma_m\}$ be the set of all k -dimensional simplices in K . Then, an element of $C_k(K)$ is a formal linear combination:

$$\sum_{j=1}^m n_j \sigma_j, \quad n_j \in \mathbb{F}_2.$$

Then, $C_k(K)$ is a vector space over $\mathbb{F}_2$ with $K^{(k)}$ as a basis.

- Boundary operator:

$$\begin{aligned} \partial_k : C_k(K) &\rightarrow C_{k-1}(K) \\ \sigma = [v_0, \dots, v_k] &\mapsto \sum_{j=1}^k [v_0, \dots, \hat{v}_j, \dots, v_k] \end{aligned}$$

where $\hat{v}_j$ indicates that v_j is omitted.

- Then, $\partial_k \left(\sum_{j=1}^m n_j \sigma_j \right) := \sum_{j=1}^m n_j \partial_k \sigma_j$, i.e. ∂_k is a linear operator.
- $\partial_1[a, b] = [b] + [a]$, $\partial_2[a, b, c] = [b, c] + [a, c] + [a, b]$, $\partial_2([a, b, c] + [d, e, f]) = \partial_2[a, b, c] + \partial_2[d, e, f]$
- $\beta_k(K) = \dim(H_k(K)) = \text{nullity}(\partial_k) - \text{rank}(\partial_{k+1})$, $k \in \{0, \dots, d\}$

Vietoris-Rips Complex

Let $X \subset M \subset \mathbb{R}$ be a point-cloud data set. Then,

$$\text{VR}_\varepsilon(X) := \{[x_0, \dots, x_k] : \delta(x_i, x_j) \leq 2\varepsilon, x_0, \dots, x_k \in X, k = 0, 1, \dots, n\},$$

i.e. the set of k -simplices such that to vertices are contained in a non-zero simplex if and only if they are within the prescribed distance to each other.

We need to choose ε and δ to create the simplicial-complex with the most information possible, without overfitting the dataset to a discrete set ($\varepsilon = 0$).

- δ is chosen as the geodesic distance in M , estimated from X using the graph geodesic distance
- to obtain ε_* , we use *persistent barcodes*

Persistent Homology

- Generally speaking, a persistence barcode is an interval $[\varepsilon, \varepsilon')$ where its left-end point ε is the scale at which the new feature appears or born, and its right-end point ε' is the scale at which that feature disappears or die. The length of the interval $\varepsilon' - \varepsilon$ is the persistence of that feature.
- Features that are non-robust to perturbations will produce short intervals; conversely, features that persist long enough, i.e., produce long intervals, are thought to be prominent features of the underlying manifold
- We are interested in selecting ε_* from a finite set of $\varepsilon_0 < \varepsilon_1 < \dots < \varepsilon_m$. Letting $K_j = VR_{\varepsilon_j}(X)$, we then have that $K_0 \subset K_1 \subset \dots \subset K_m$.

$$\begin{array}{cccccccccccccccccccc}
0 & \xrightarrow{\partial_{d+1}} & C_d(K_1) & \xrightarrow{\partial_d} & C_{d-1}(K_1) & \xrightarrow{\partial_{d-1}} & \dots & \xrightarrow{\partial_{k+1}} & C_k(K_1) & \xrightarrow{\partial_k} & C_{k-1}(K_1) & \xrightarrow{\partial_{k+1}} & \dots & \xrightarrow{\partial_2} & C_1(K_1) & \xrightarrow{\partial_1} & C_0(K_1) & \xrightarrow{\partial_0} & 0 \\
& & \downarrow f_1 & & \downarrow f_1 & & & & \downarrow f_1 & & \downarrow f_1 & & & & \downarrow f_1 & & \downarrow f_1 & & \\
0 & \xrightarrow{\partial_{d+1}} & C_d(K_2) & \xrightarrow{\partial_d} & C_{d-1}(K_2) & \xrightarrow{\partial_{d-1}} & \dots & \xrightarrow{\partial_{k+1}} & C_k(K_2) & \xrightarrow{\partial_k} & C_{k-1}(K_2) & \xrightarrow{\partial_{k+1}} & \dots & \xrightarrow{\partial_2} & C_1(K_2) & \xrightarrow{\partial_1} & C_0(K_2) & \xrightarrow{\partial_0} & 0 \\
& & \downarrow f_2 & & \downarrow f_2 & & & & \downarrow f_2 & & \downarrow f_2 & & & & \downarrow f_2 & & \downarrow f_2 & & \\
0 & \xrightarrow{\partial_{d+1}} & C_d(K_3) & \xrightarrow{\partial_d} & C_{d-1}(K_3) & \xrightarrow{\partial_{d-1}} & \dots & \xrightarrow{\partial_{k+1}} & C_k(K_3) & \xrightarrow{\partial_k} & C_{k-1}(K_3) & \xrightarrow{\partial_{k+1}} & \dots & \xrightarrow{\partial_2} & C_1(K_3) & \xrightarrow{\partial_1} & C_0(K_3) & \xrightarrow{\partial_0} & 0 \\
& & \downarrow f_3 & & \downarrow f_3 & & & & \downarrow f_3 & & \downarrow f_3 & & & & \downarrow f_3 & & \downarrow f_3 & & \\
\vdots & & \vdots & & \vdots & & \vdots & & \vdots & & \vdots & & \vdots & & \vdots & & \vdots & & \vdots \\
& & \downarrow f_{m-1} & & \downarrow f_{m-1} & & & & \downarrow f_{m-1} & & \downarrow f_{m-1} & & & & \downarrow f_{m-1} & & \downarrow f_{m-1} & & \\
0 & \xrightarrow{\partial_{d+1}} & C_d(K_m) & \xrightarrow{\partial_d} & C_{d-1}(K_m) & \xrightarrow{\partial_{d-1}} & \dots & \xrightarrow{\partial_{k+1}} & C_k(K_m) & \xrightarrow{\partial_k} & C_{k-1}(K_m) & \xrightarrow{\partial_{k+1}} & \dots & \xrightarrow{\partial_2} & C_1(K_m) & \xrightarrow{\partial_1} & C_0(K_m) & \xrightarrow{\partial_0} & 0
\end{array}$$

Planning

- Visualization: write a Python script to visualize a data set over an underlying manifold and allow the user to select an ϵ value for the Vietoris-Rips complex. Should have a 3d visual.